

The empirical recurrence for OEIS A299510, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A299510 records “Empirical recurrence of order 68” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 71$. Its last coefficient is $c_{68} = -16 \neq 0$, so no further tail satisfies anything shorter; any order-68 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A299510 is “Number of $n \times 4$ 0..1 arrays with every element equal to 1, 2, 4, 5, 6 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

2, 13, 29, 112, 439, 1950, 7702, 30277, ...

The entry states:

Empirical recurrence of order 68 (see link above).

As of the “Last modified” line on the live entry (Jan 20 2024 16:16:20, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 68: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 68.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 3$ on, it returns order 68 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{68} c_i a(n-i),$$

c_1	8	c_{18}	−1627915	c_{35}	−90206512	c_{52}	23955963
c_2	−25	c_{19}	2325617	c_{36}	12480940	c_{53}	−6925319
c_3	62	c_{20}	−3483380	c_{37}	−57814053	c_{54}	−19478866
c_4	−146	c_{21}	5757152	c_{38}	151156061	c_{55}	8221526
c_5	251	c_{22}	−7758224	c_{39}	−15674484	c_{56}	7203952
c_6	−435	c_{23}	9178993	c_{40}	71907016	c_{57}	1647264
c_7	840	c_{24}	−9319242	c_{41}	−178929826	c_{58}	−4790372
c_8	−2152	c_{25}	11804939	c_{42}	140884859	c_{59}	−1173742
c_9	4550	c_{26}	−13847525	c_{43}	−102110415	c_{60}	584993
c_{10}	−6325	c_{27}	8517624	c_{44}	120676781	c_{61}	419256
c_{11}	13683	c_{28}	−5842294	c_{45}	−178595845	c_{62}	12120
c_{12}	−36097	c_{29}	3000931	c_{46}	96305791	c_{63}	−116744
c_{13}	82061	c_{30}	15189375	c_{47}	−35768143	c_{64}	24017
c_{14}	−123284	c_{31}	−30439935	c_{48}	52114630	c_{65}	699
c_{15}	166729	c_{32}	31725392	c_{49}	−69249776	c_{66}	1270
c_{16}	−378488	c_{33}	−42928948	c_{50}	−4464831	c_{67}	312
c_{17}	855276	c_{34}	71914954	c_{51}	31720481	c_{68}	−16

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 71$, and it is the recurrence OEIS A299510 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{68} - c_1 x^{68-1} - \dots - c_{68}$. Its constant term is $-c_{68} = 16$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 68 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 68, so $\deg q = 68$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 68, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 23 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 68, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -16 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-68 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A299510.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.